FARMLINK DATABASE – ALL TABLES

Normalized Relational Architecture (3NF) • MySQL Schema • Prefixed Primary Keys

1. USERS

Farmer / Buyer / Admin

FIELD	DATA TYPE	KEY	NULL	DEFAULT	DESCRIPTION
u_id	INT	PK	NO	AUTO_INC	User unique ID
name	VARCHAR(100)	-	NO	-	User full name
email	VARCHAR(255)	UNIQUE	NO	-	Login email address
password_hash	VARCHAR(255)	-	NO	-	Bcrypt password hash
account_type	ENUM('farmer','business','admin')	-	NO	-	User role & access
created_at	TIMESTAMP	-	YES	CURRENT_TIMESTAMP	Account registration time

💡 **Role Model:** Farmers, Buyers, and Admins are authenticated via `account_type`.

2. PRODUCTS

Produce Catalog & Wholesale

FIELD	DATA TYPE	KEY	NULL	DEFAULT	DESCRIPTION
p_id	INT	PK	NO	AUTO_INC	Product unique ID
farmer_id	INT	FK	NO	-	Seller (users.u_id)
name	VARCHAR(150)	-	NO	-	Produce item name
category	VARCHAR(50)	-	NO	-	Vegetables, Fruits, Grains
quantity	DECIMAL(10,2)	-	NO	-	Available stock inventory
unit	ENUM('kg','quintal','bag','crate')	-	NO	-	Produce unit measurement
price	DECIMAL(10,2)	-	NO	-	Base unit price (₹)
location	VARCHAR(150)	-	NO	-	Farm / City location
image_url	VARCHAR(500)	-	YES	NULL	Produce photo URL
tier1_min_qty	INT	-	NO	51	Wholesale tier 1 min qty
tier1_discount_pct	INT	-	NO	10	Wholesale tier 1 discount %
tier2_min_qty	INT	-	NO	201	Wholesale tier 2 min qty
tier2_discount_pct	INT	-	NO	20	Wholesale tier 2 discount %
created_at	TIMESTAMP	-	YES	CURRENT_TIMESTAMP	Created timestamp
updated_at	TIMESTAMP	-	YES	CURRENT_TIMESTAMP	Last updated timestamp

🔑 **FK:** `farmer_id` ➔ `users.u_id`

3. DELIVERY_ADDRESSES

Buyer Addresses

FIELD	DATA TYPE	KEY	NULL	DEFAULT	DESCRIPTION
da_id	INT	PK	NO	AUTO_INC	Address unique ID
business_id	INT	FK	NO	-	Buyer (users.u_id)
recipient_name	VARCHAR(100)	-	NO	-	Receiver person name
phone	VARCHAR(20)	-	NO	-	Contact phone number
address_line1	VARCHAR(255)	-	NO	-	Street / Building address
address_line2	VARCHAR(255)	-	YES	NULL	Landmark / Area
city	VARCHAR(100)	-	NO	-	Delivery city (Gujarat)
state	VARCHAR(100)	-	NO	Gujarat	State (Gujarat)
pincode	VARCHAR(10)	-	NO	-	Postal 6-digit code
is_default	TINYINT(1)	-	NO	0	Default address flag
created_at	TIMESTAMP	-	YES	CURRENT_TIMESTAMP	Address created time

🔑 FK: business_id ➔ users.u_id

4. ORDERS

Master Checkout Orders

FIELD	DATA TYPE	KEY	NULL
o_id	INT	PK	NO
business_id	INT	FK	NO
delivery_address_id	INT	FK	YES
total	DECIMAL(12,2)	-	NO
status	ENUM('Awaiting Payment','Pending','Accepted','Out for Delivery','Delivered','Cancelled','CANCELLED_BY_BUYER')	-	YES
tracking_code	VARCHAR(100)	-	YES
delivery_otp	VARCHAR(10)	-	YES
delivery_name	VARCHAR(100)	-	YES
delivery_phone	VARCHAR(20)	-	YES
delivery_address	TEXT	-	YES
payment_method	ENUM('COD','ONLINE')	-	NO
payment_status	ENUM('NOT_REQUIRED','PENDING','PAID','FAILED','REFUND_PENDING','REFUNDED')	-	NO
gateway_order_id	VARCHAR(100)	-	YES
gateway_payment_id	VARCHAR(100)	-	YES
cancelled_by_user_id	BIGINT UNSIGNED	-	YES
cancelled_by_role	VARCHAR(20)	-	YES
cancellation_reason	VARCHAR(255)	-	YES
cancelled_at	TIMESTAMP	-	YES
created_at	TIMESTAMP	-	YES

🔑 **FKs:** business_id ➔ users.u_id, delivery_address_id ➔ delivery_addresses.da_id

5. ORDER_ITEMS

Produce Items in Order

FIELD	DATA TYPE	KEY	NULL	DEFAULT	DE
oi_id	BIGINT UNSIGNED	PK	NO	AUTO_INC	Or
order_id	INT	FK	NO	-	or
product_id	INT	FK	NO	-	pr
farmer_id	INT	FK	NO	-	Fa (u
quantity	DECIMAL(10,2)	-	NO	-	Pu qu
unit_price	DECIMAL(10,2)	-	NO	-	Pr
discount_amount	DECIMAL(10,2)	-	NO	0.00	W sa
discount_pct	INT	-	NO	0	W di
original_total	DECIMAL(10,2)	-	YES	NULL	Pr di
total	DECIMAL(10,2)	-	NO	-	Ne to
status	ENUM('Awaiting Payment','Pending','Accepted','Out for Delivery','Delivered','Cancelled','CANCELLED_BY_BUYER')	-	YES	Pending	Ite
created_at	TIMESTAMP	-	NO	CURRENT_TIMESTAMP	Ite tir

🔑 **FKs:** order_id ➔ orders.o_id, product_id ➔ products.p_id, farmer_id ➔ users.u_id

6. PAYMENTS

Payment Transactions

FIELD	DATA TYPE	KEY	NULL	DEFAULT	DESCRIPTION
pay_id	BIGINT UNSIGNED	PK	NO	AUTO_INC	Payment record unique ID
order_id	INT	FK	NO	-	orders.o_id
amount	DECIMAL(12,2)	-	NO	-	Paid / Checkout amount
payment_method	ENUM('COD','ONLINE')	-	NO	COD	Cash on Delivery / Online
payment_status	ENUM('PENDING','PAID','FAILED','REFUNDED')	-	NO	PENDING	Gateway payment status
gateway_order_id	VARCHAR(100)	-	YES	NULL	Razorpay order reference
gateway_payment_id	VARCHAR(100)	-	YES	NULL	Razorpay capture reference
created_at	TIMESTAMP	-	NO	CURRENT_TIMESTAMP	Payment initiated time
updated_at	TIMESTAMP	-	NO	CURRENT_TIMESTAMP	Payment updated time

🔑 FK: order_id ➔ orders.o_id

7. ORDER_TRACKING_EVENTS

Status Timeline Logs

FIELD	DATA TYPE	KEY	NULL	DEFAULT	DESCRIPTION
ote_id	BIGINT UNSIGNED	PK	NO	AUTO_INC	Tracking event ID
order_id	INT	FK	NO	-	orders.o_id
status	VARCHAR(60)	-	NO	-	Order status milestone
note	VARCHAR(255)	-	NO	-	Status update explanation
created_at	TIMESTAMP	-	NO	CURRENT_TIMESTAMP	Milestone timestamp

🔑 FK: order_id ➔ orders.o_id

8. ORDER_DELIVERY_LOCATIONS

Live Driver GPS Locations

FIELD	DATA TYPE	KEY	NULL	DEFAULT	DESCRIPTION
odl_id	BIGINT UNSIGNED	PK	NO	AUTO_INC	Location coordinate ID
order_id	INT	FK	NO	-	orders.o_id
latitude	DECIMAL(10,7)	-	NO	-	GPS Latitude coordinate
longitude	DECIMAL(10,7)	-	NO	-	GPS Longitude coordinate
note	VARCHAR(255)	-	NO	-	Delivery status note
created_at	TIMESTAMP	-	NO	CURRENT_TIMESTAMP	Coordinate logged timestamp

🔑 **FK:** `order_id` ➔ `orders.o_id`

9. ORDER_NOTIFICATIONS

In-App Alerts (with Dismiss)

FIELD	DATA TYPE	KEY	NULL	DEFAULT	DESCRIPTION
on_id	BIGINT UNSIGNED	PK	NO	AUTO_INC	Notification ID
user_id	BIGINT UNSIGNED	FK	NO	-	Recipient (users.u_id)
order_id	BIGINT UNSIGNED	FK	NO	-	orders.o_id
type	VARCHAR(80)	-	YES	NULL	ORDER_DELIVERED, etc.
title	VARCHAR(120)	-	YES	NULL	Notification header
message	VARCHAR(500)	-	NO	-	Alert body message
is_read	TINYINT(1)	-	NO	0	Read / Unread flag
created_at	TIMESTAMP	-	NO	CURRENT_TIMESTAMP	Notification timestamp

🔑 **FKs:** `user_id` ➔ `users.u_id`, `order_id` ➔ `orders.o_id`

FOREIGN KEY RELATIONSHIPS

CHILD TABLE	FOREIGN KEY	REFERENCES TABLE	PARENT KEY
products	farmer_id	users	u_id
delivery_addresses	business_id	users	u_id
orders	business_id	users	u_id
orders	delivery_address_id	delivery_addresses	da_id
order_items	order_id	orders	o_id
order_items	product_id	products	p_id
order_items	farmer_id	users	u_id

SCHEMA LEGEND & CONVENTIONS

PK Primary Key (Unique Table ID)

FK Foreign Key (Relational Link)

UNIQUE Unique Constraint

NO Not Null (Mandatory)

YES Can be NULL (Optional)

Design Principles Applied:

- **3NF Normalization:** Zero redundant columns.

CHILD TABLE	FOREIGN KEY	REFERENCES TABLE	PARENT KEY
payments	`order_id`	orders	`o_id`
order_tracking_events	`order_id`	orders	`o_id`
order_delivery_locations	`order_id`	orders	`o_id`
order_notifications	`user_id`	users	`u_id`
order_notifications	`order_id`	orders	`o_id`

- **Prefixed Primary Keys:** Easy recognition in MySQL Workbench.
- **Dedicated Payments:** 1-to-1 relation with Orders.